

Cloud Data Management: An Overview

Organisations need to build a strategy for managing the data they send to the cloud, since data security and privacy are important. They must also choose the cloud service provider carefully.



Cloud data management is all about the end-to-end process of managing data in the cloud, right from collection to analytics. It includes managing the type, volume and nature of data in a cloud environment. When organisations use public cloud environments for their enterprise data, they need to have a proper strategy since the data is stored in an external data centre. Public clouds are inherently multi-tenant; hence, security and privacy of the data is important. A good data strategy addresses data in transit and data at rest.

Key aspects of data management in the cloud

Data management in the cloud offers the following benefits.

- **Collection/Ingestion**
 - Availability of connectors/APIs that can import data from a variety of sources (databases, files, real-time data, etc). This also includes cleaning the data and handling missing data.
- **Integration and transformation**
 - Ability to map the data and integrate with other data.
- **Storage**
 - Ability to store data in the right format in an efficient manner.
- **Retrieval**
 - Users are able to read/retrieve the data.
- **Security**
 - Ability to process data in a secured manner for both data at rest and data in transit
- **Privacy**
 - Ability to mask sensitive data.
- **Backup and recovery**
 - Ability to provide automated backups and recovery.
- **Metadata management**
 - Providing good meta data to understand the characteristics of the data stored in the cloud.